MPCA Project

TITLE OF THE PROJECT: SMART AIR POLLUTION MONITORING SYSTEM

Team Members:

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Components used:

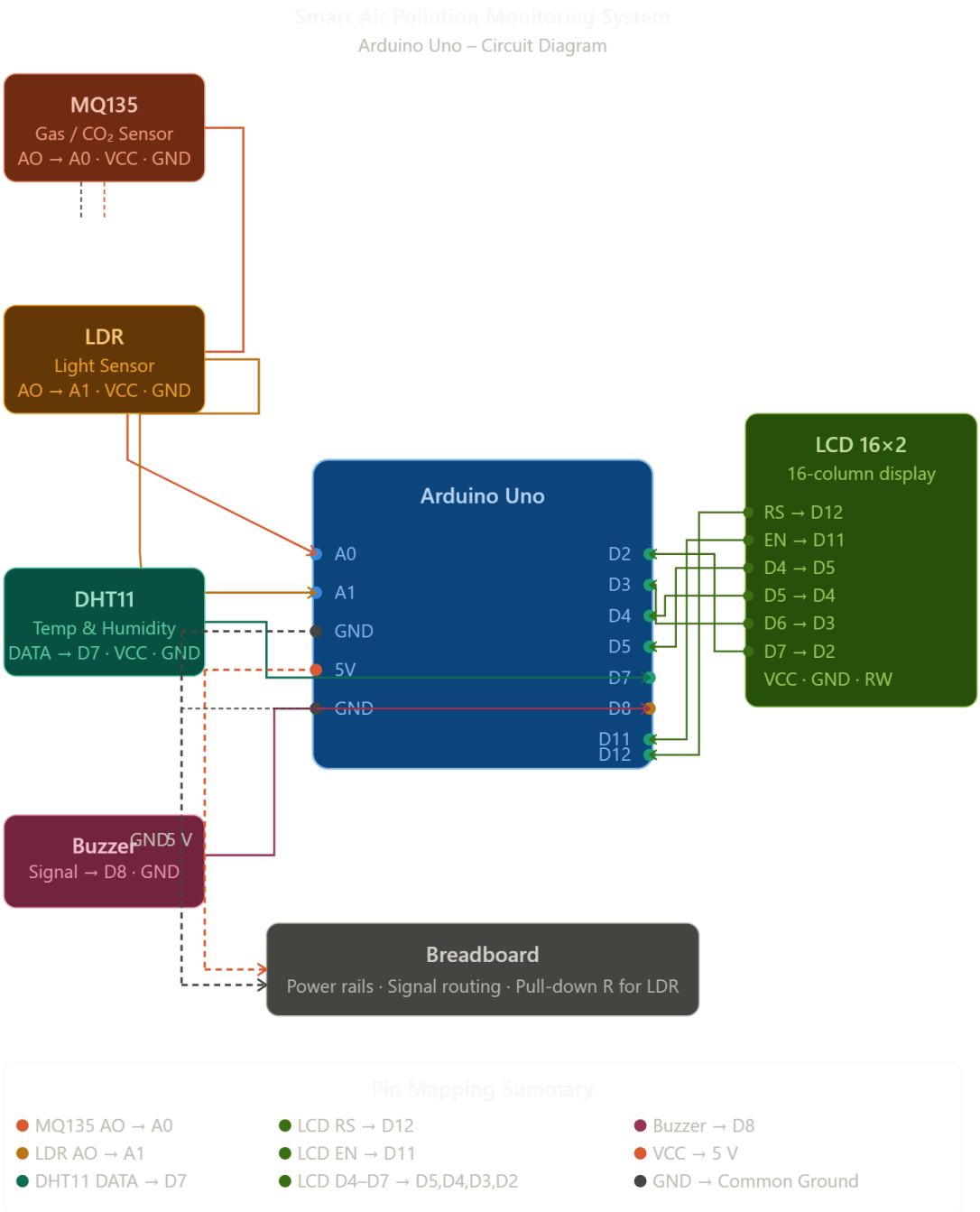
- Arduino Uno (Microcontroller)
- MQ135 Gas Sensor
- DHT11 Temperature and Humidity Sensor
- LDR Sensor
- 16x2 LCD Display
- Buzzer
- Breadboard
- Jumper Wires
- Power Supply (USB)

Explanation about the project :

The Smart Air Pollution Monitoring System is designed to measure and monitor air quality in real time using multiple sensors. The MQ135 sensor detects harmful gases such as carbon dioxide, smoke, and ammonia present in the air. The DHT11 sensor measures temperature and humidity, while the LDR sensor monitors light intensity.

All sensor data is collected and processed by the Arduino microcontroller. The processed values are displayed on a 16x2 LCD screen, providing real-time air quality information. If the pollution level exceeds a predefined threshold, a buzzer is activated to alert the user.

Circuit Diagram:



Pseudocode:

```
Start

Initialize LCD, sensors, and buzzer

Repeat:
   Read gas sensor value
   Read temperature and humidity
   Read light intensity

   Display values on LCD

   If gas value > threshold:
      Turn ON buzzer
   Else:
      Turn OFF buzzer

End Repeat
```

Output Pictures with explanations:

Explanation of Output :

In this output, the Smart Air Pollution Monitoring System displays real-time environmental data on the LCD screen.

- Gas: 156 → This value is obtained from the MQ135 gas sensor. It indicates the concentration of harmful gases in the air. A value of 156 represents a moderate level of pollution.
- Light: Bright → This is derived from the LDR sensor. It shows that the surrounding environment has high light intensity.

The system processes sensor data using the Arduino microcontroller and updates the LCD continuously.
The red LED is glowing, which indicates that the gas level has crossed a predefined threshold and the system is alerting the user about air quality conditions.

Explanation of Output :

In this output, the Smart Air Pollution Monitoring System displays environmental parameters measured using the DHT11 sensor.

- **Temp: 33.60°C** → This is the temperature of the surrounding environment. It indicates that the environment is relatively warm.
- **Hum: 28.00%** → This shows the humidity level in the air, which is relatively low, indicating dry conditions.

The Arduino microcontroller reads data from the DHT11 sensor and continuously updates the LCD display with real-time values.
The **red LED is glowing**, indicating that the system is active and may also serve as an alert indicator based on predefined conditions